

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for **Admitix**, a cloud-based AI-powered Admission Workflow Management System. The system aims to simplify and automate the post-allotment admission process for higher education institutions.

1.2 Scope

Admitix is a multi-tenant SaaS platform that enables colleges to manage admissions digitally through configurable workflows, AI-assisted document verification, approval management, and student record creation.

The system is designed for:

- Engineering Colleges
- Polytechnic Colleges
- Medical Colleges
- Pharmacy Colleges
- Law Colleges

1.3 Intended Users

- Super Admin
- Institution Admin
- Admission Officer
- Department Reviewer
- Finance Officer
- Registrar / Student Records Officer

- Student
- Parent / Guardian (Optional)

2. Overall Description

Product Perspective

Admitix is an independent cloud platform designed to streamline post-allotment admissions while allowing future integration with existing ERP systems.

Product Objectives

- Digitize admissions
- Reduce manual paperwork
- Improve transparency
- Automate document verification
- Simplify approval workflow
- Support multiple institutions

3. Functional Requirements

3.1 Authentication Module

- Secure Login
- Password Reset
- Multi-factor Authentication (Future)
- Role-Based Access Control (RBAC)

3.2 Institution Management

- Institution Registration

- Department Management
- Course Management
- Academic Year Management
- Admission Cycle Configuration

3.3 Student Management

- Student Registration
- Student Profile
- Admission Application
- Admission Status
- Student Record Generation

3.4 Document Management

- Document Upload
- Document Preview
- Document Verification
- Document Approval
- Document Re-upload Request
- Document Storage

3.5 Admission Workflow

- Application Review
- Approval Workflow
- Admission Tracking
- Admission Confirmation
- Admission History

3.6 AI Module

- OCR Data Extraction

- Missing Document Detection
- Cross-document Validation
- Duplicate Detection
- Blur Detection
- AI Admission Assistant

3.7 Finance Module

- Fee Status
- Payment Verification
- Receipt Generation
- Scholarship Status (Future)

3.8 Reporting Module

- Admission Reports
- Pending Applications
- Completed Admissions
- Department Reports
- Student Statistics

3.9 Notification Module

- Email Notifications
- SMS Notifications (Future)
- In-App Notifications

3.10 Super Admin Module

- Manage Colleges
- Subscription Management
- Platform Monitoring
- Analytics Dashboard

- System Configuration

4. Non-Functional Requirements

Performance

- Support multiple colleges simultaneously.
- Fast response time.
- High availability.

Security

- Role-Based Access Control
- Password Encryption
- HTTPS Communication
- Audit Logs
- Secure Cloud Storage

Scalability

- Multi-Tenant Architecture
- Horizontal Scaling
- Modular Design

Reliability

- Daily Backup
- Error Logging
- Disaster Recovery Support

Usability

- Responsive Interface
- Simple Navigation
- Mobile Friendly

Maintainability

- Modular Architecture
- Easy Updates
- API Support

5. External Interface Requirements

User Interface

- Web-based dashboard
- Responsive design
- Role-specific dashboard
- Dark/Light Mode (Future)

Software Interfaces

Future Integration:

- ERP Systems
- University Portals
- Payment Gateway
- Email Services

- SMS Gateway

Hardware Interfaces

- Desktop
- Laptop
- Tablet
- Mobile Browser

6. Database Requirements

Main entities include:

- Institution
- Department
- Course
- Academic Year
- Student
- Admission Application
- Documents
- Workflow
- Users
- Roles
- Permissions
- Notifications
- Audit Logs

7. Security Requirements

- Secure Authentication
- RBAC
- Data Encryption

- Audit Trail
- Session Management
- Backup & Recovery

8. Future Enhancements

- ERP Integration
- University Portal Integration
- Mobile Application
- AI Copilot
- AI Recommendation Engine
- Predictive Analytics
- Digital Approval Workflow
- Digital Signature Support
- API Marketplace

9. Assumptions & Constraints

Assumptions

- Institutions have internet connectivity.
- Users possess valid login credentials.
- Students upload valid documents.
- Institution administrators configure workflows.

Constraints

- Government admission processes (CAP/NEET/CLAT, etc.) are outside the scope.
- Accounting systems such as TallyPrime are not replaced.
- ERP integration depends on API availability.
- Institution-specific policies may vary.

10. Acceptance Criteria

The system will be considered successful if it:

- Enables colleges to manage admissions digitally.
- Supports role-based workflows.
- Successfully verifies uploaded documents.
- Maintains secure student records.
- Generates admission reports.
- Supports multiple institutions through a single SaaS platform.